

hacking HikVision IP cameras



agenda

- ~5m Introduction
- ~30m Technical Deep Dive
- ~5m Questions

whoami

[Name] Hrant
[Interests] Hardware Hacking, Reverse Engineering, Exploits...

Talks:

[ArmSec 2023] Visual Reverse Engineering
[ArmSec 2024] BIOS Malwares
[ArmSec 2024] Reverse Engineering The MacOS Sandbox
[CyberGen 2025] Reverse Engineering DJI Drones, Hardware & Software
[ArmSec 2025] Hacking HikVision IP Cameras

links

[Repository] <https://github.com/0x0000z3r0/hikvision-re>

Contact:

[Email] hrant.tadevosyan@protonmail.com

[LinkedIn] <https://www.linkedin.com/in/hrant-tadevosyan>

[Discord] 0x0000z3r0 (Z3R0)

motivation

- [+] Curiosity
- [+] Security
- [+] Modding

motivation

[+] Curiosity

[-] Security

 [*] Used everywhere in **Armenia**: banks, buildings, buses, etc.

 [*] Privacy concerns: **not** open hardware and/or source

[+] Modding

motivation

- [+] Curiosity
- [+] Security
- [-] Modding
 - [*] Extend/patch the firmware or hardware
 - [*] Fix the problems yourself

trophies

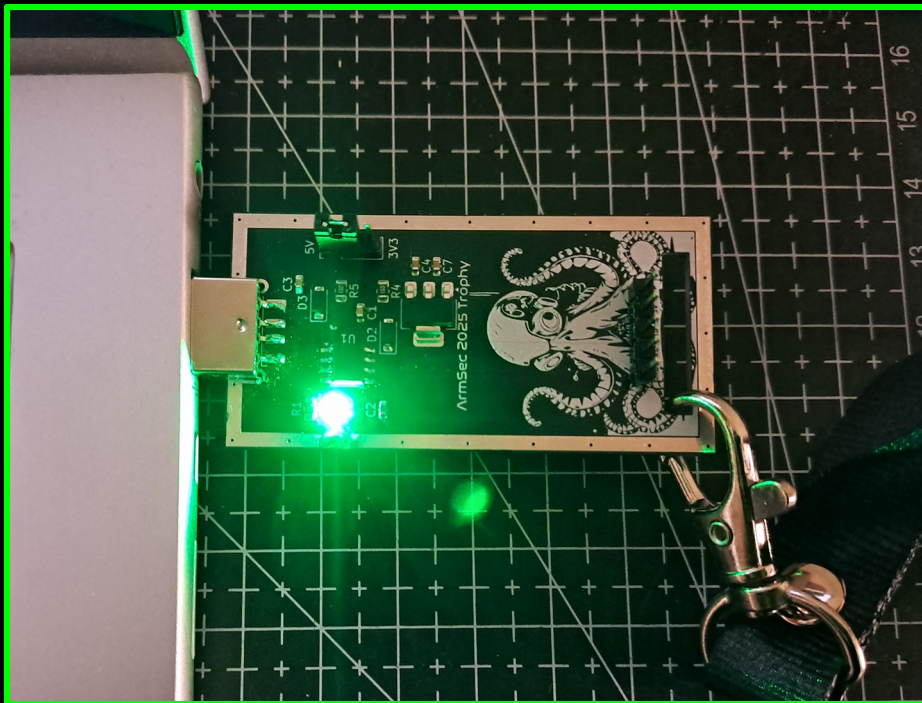
[Badges] for attentive listeners

[Difficulty] easy+



trophies (badges)

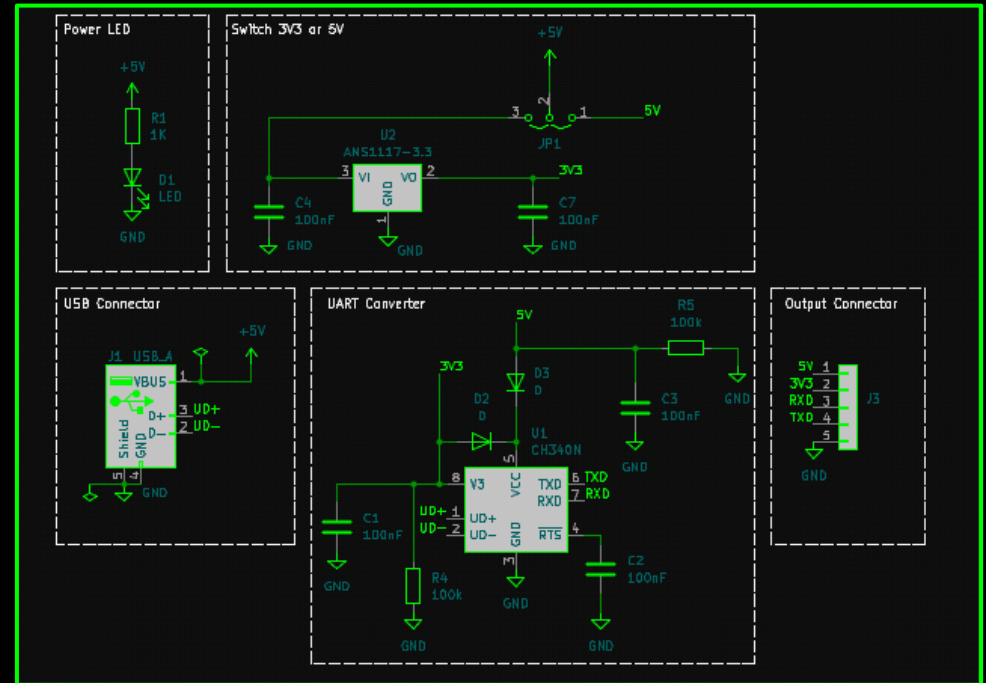
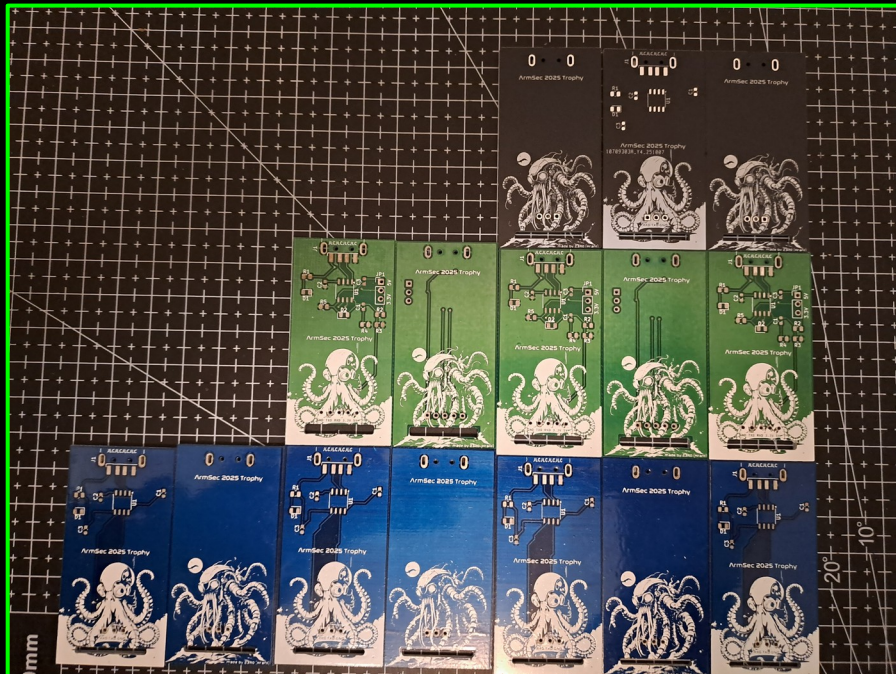
- [*] uart to usb converter
- [*] custom board design with ArmSec logo
- [*] was designed by a software person... bugs ahead



trophies (prototypes)

[*] parts need to be sourced & soldered manually

[*] btw we have a local PCB manufacturer, called **HyePeak**



clarification

[hacking] hardware hacking

[cameras] only one

...

[enviornment] casual, not a lab

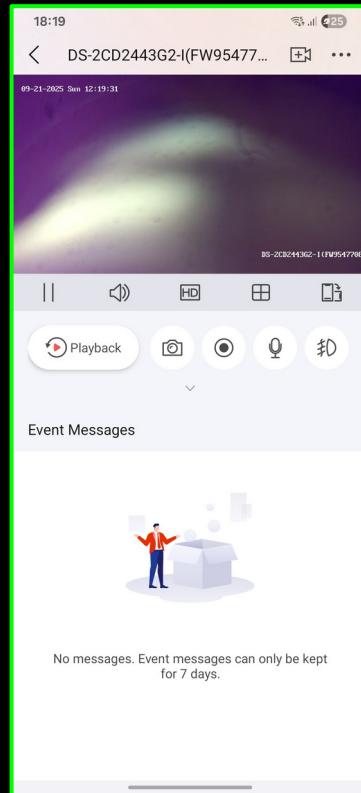
the device

[model] DS-2CD2443G2-I

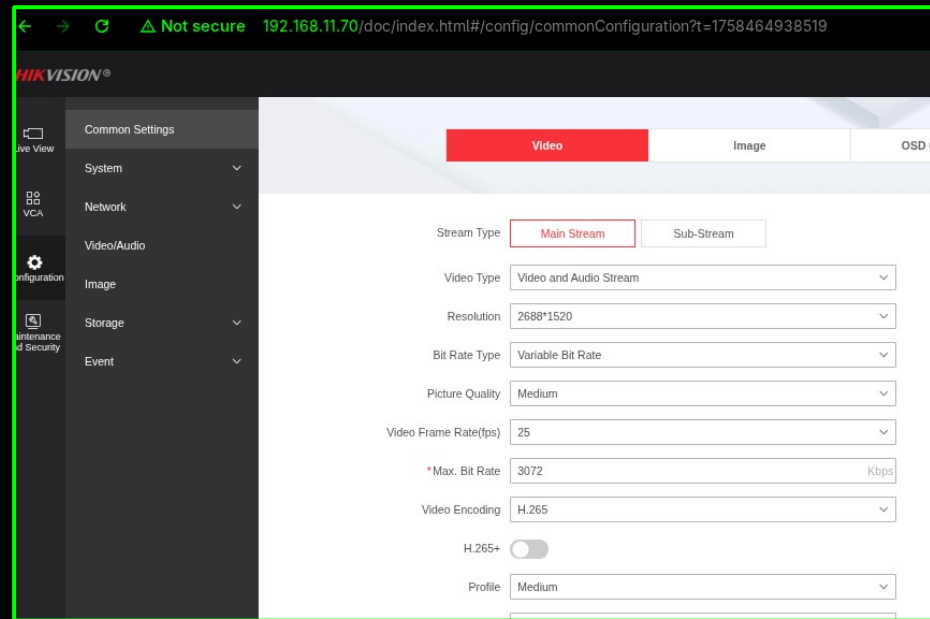
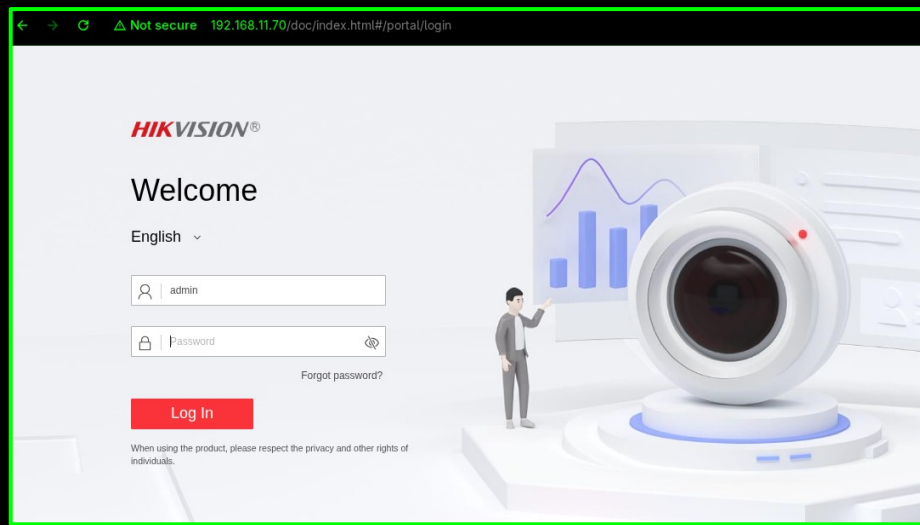
[price] ~80\$ (~100\$)



mobile UI



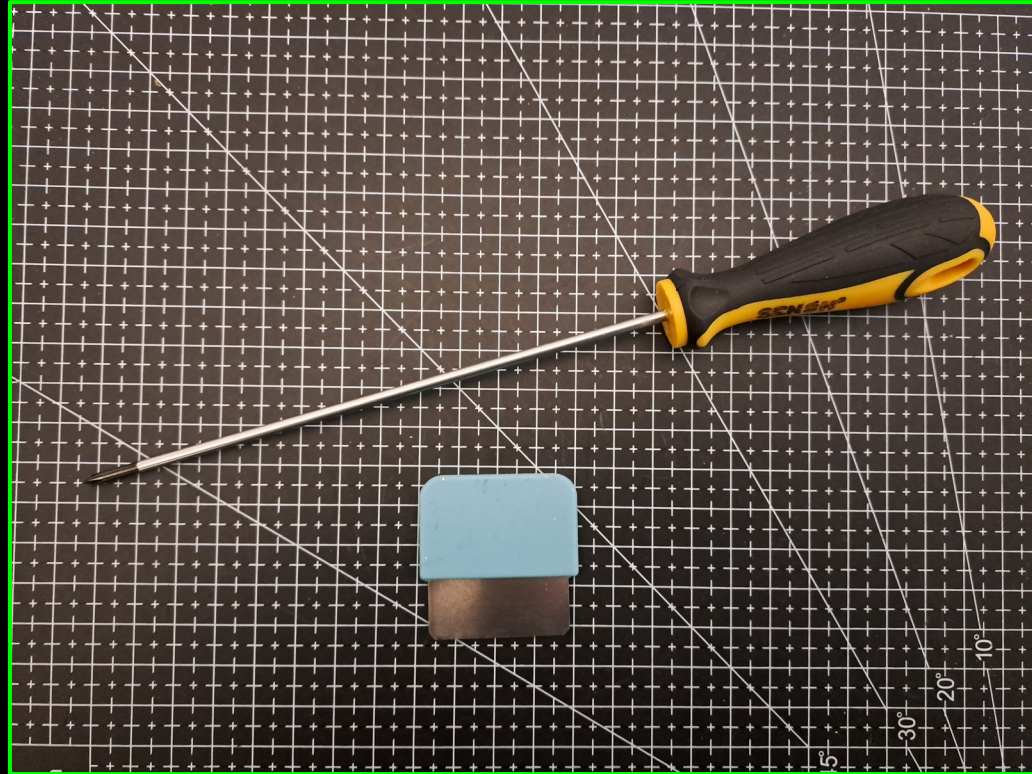
web UI



teardown tools

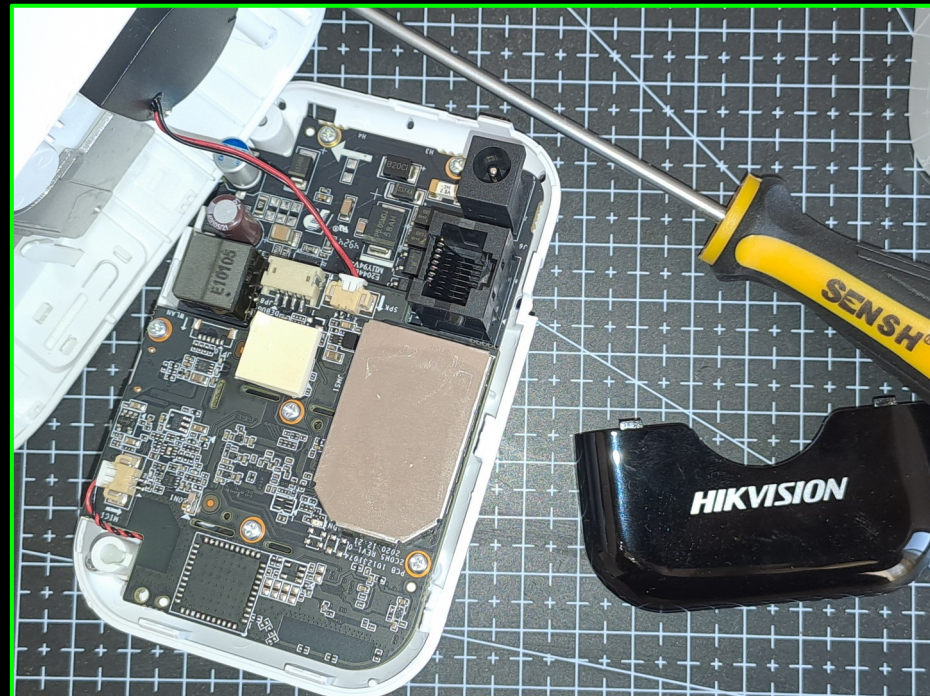
[*] PH Screwdriver

[*] Prying tool



teardown steps

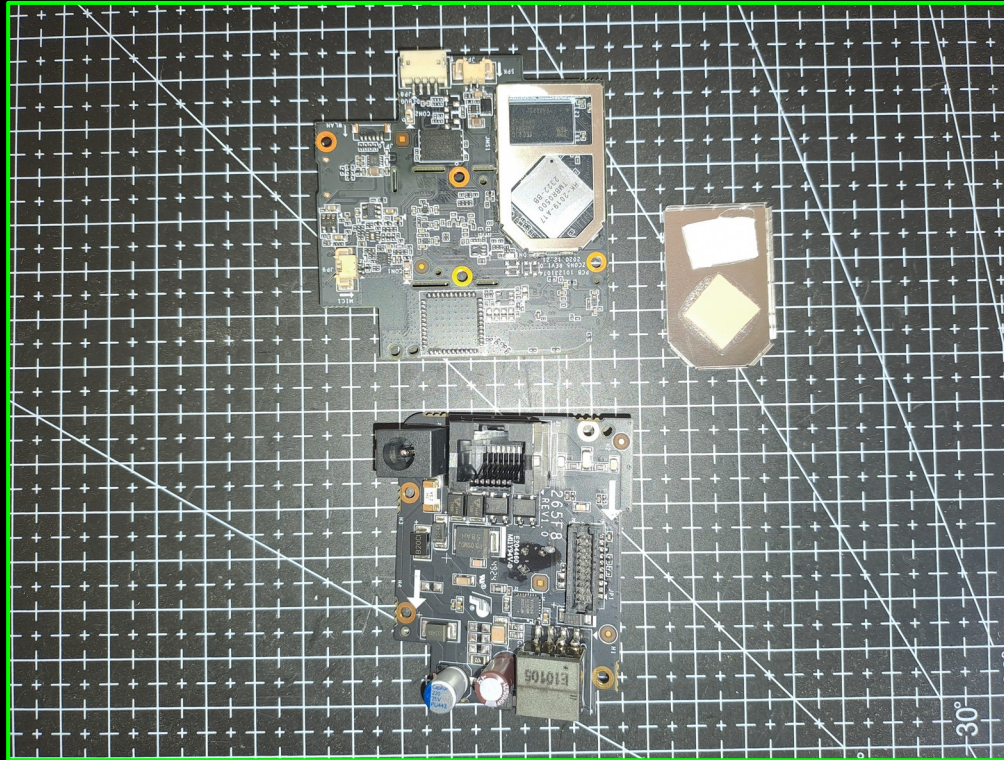
- [*] Remove the bottom housing cover
- [*] Unscrew the rest



PCBS

[*] Logic

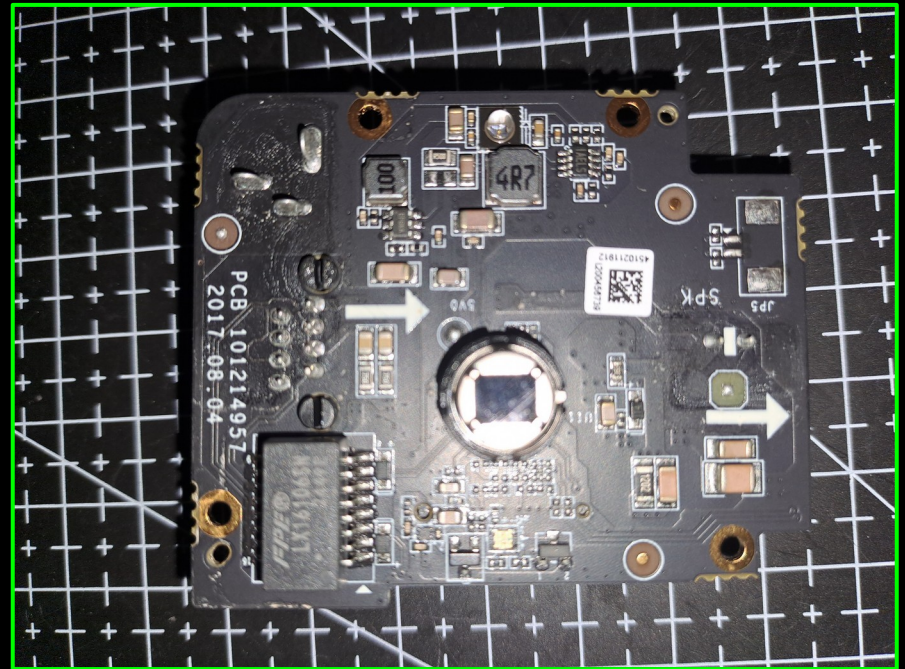
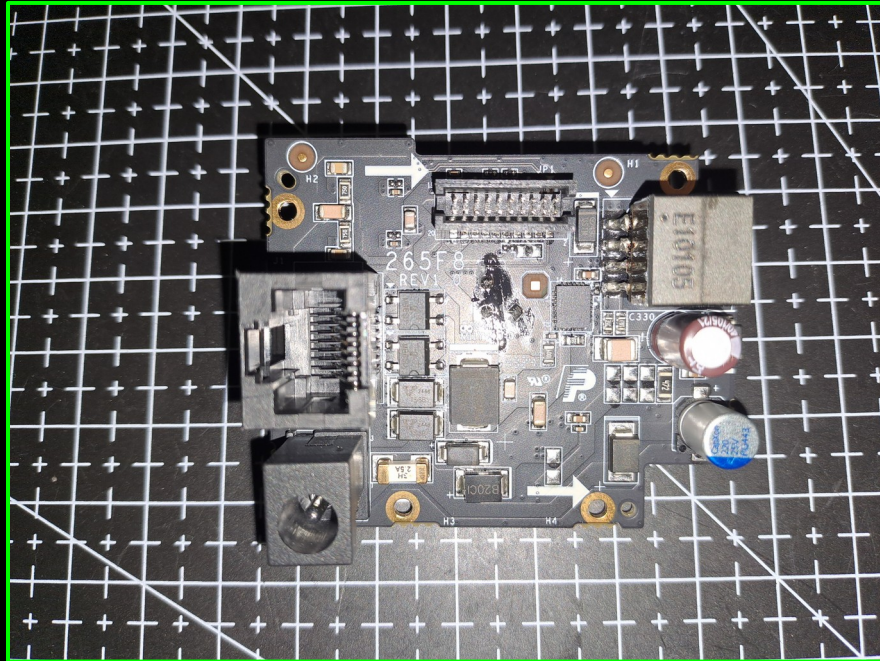
[*] Power



1st PCB

[*] Power connectors

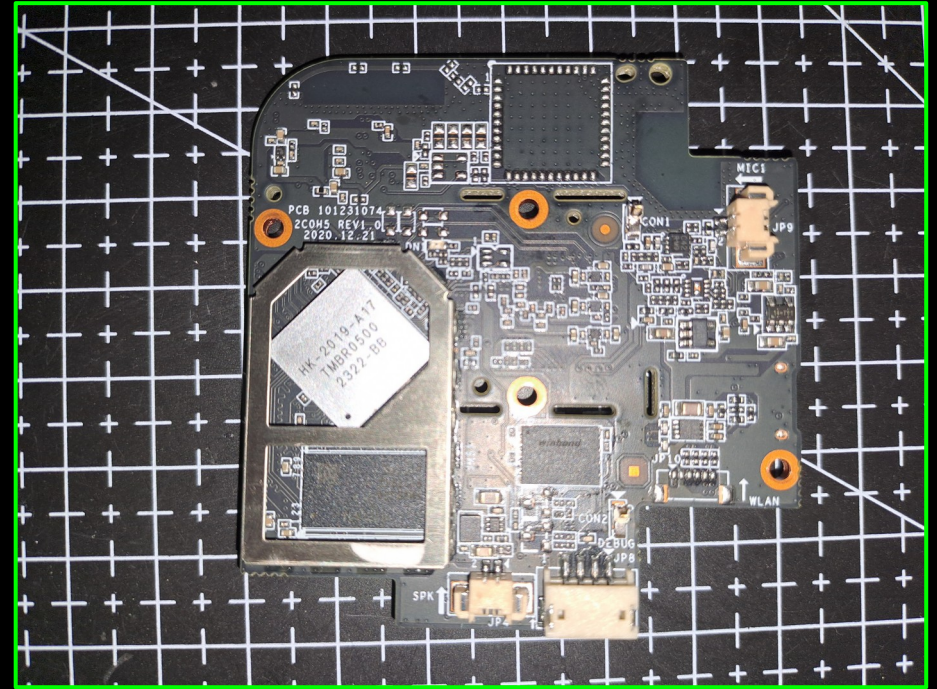
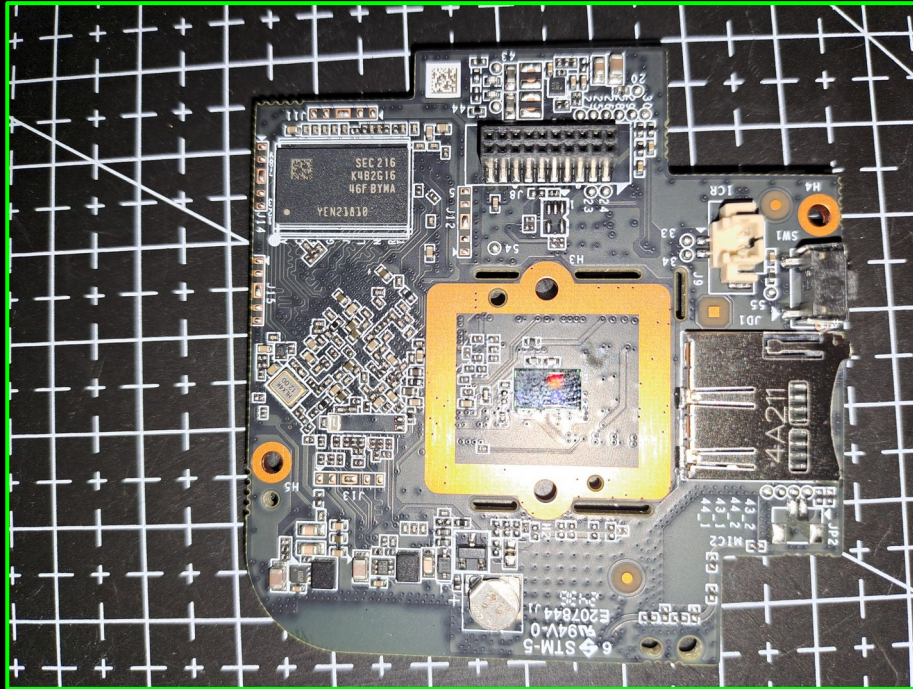
[*] Motion sensor



2nd PCB

[*] Contains the SoC

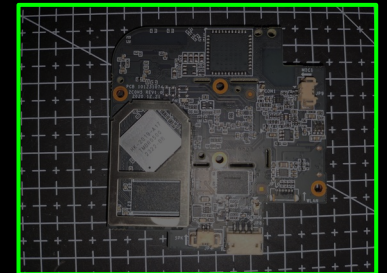
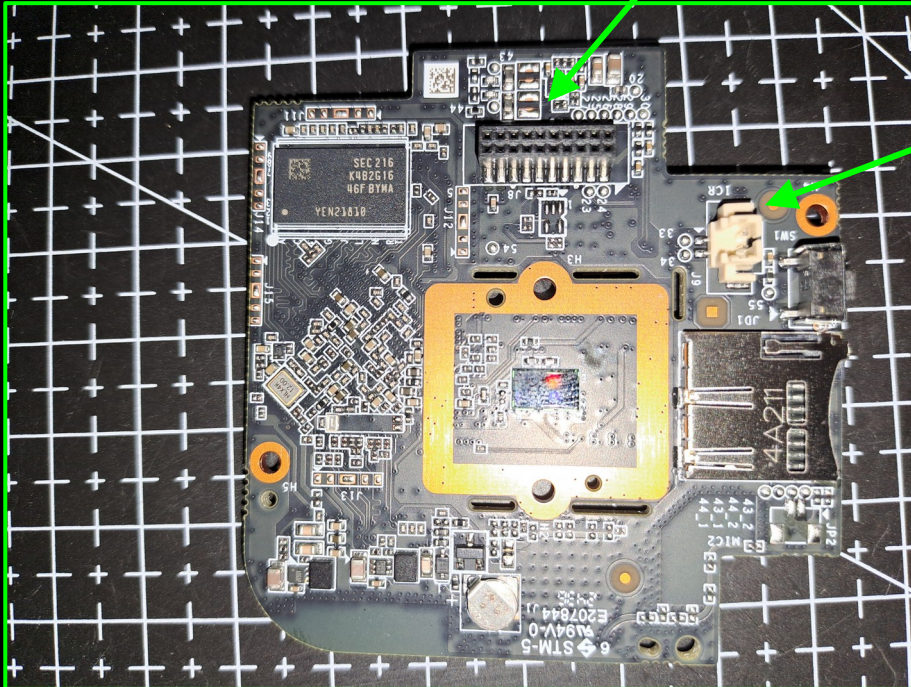
[*] ROM & RAM



connectors

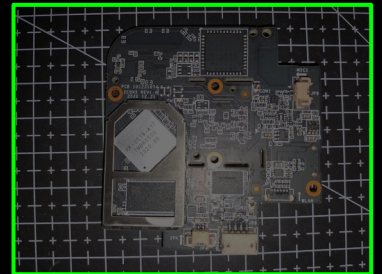
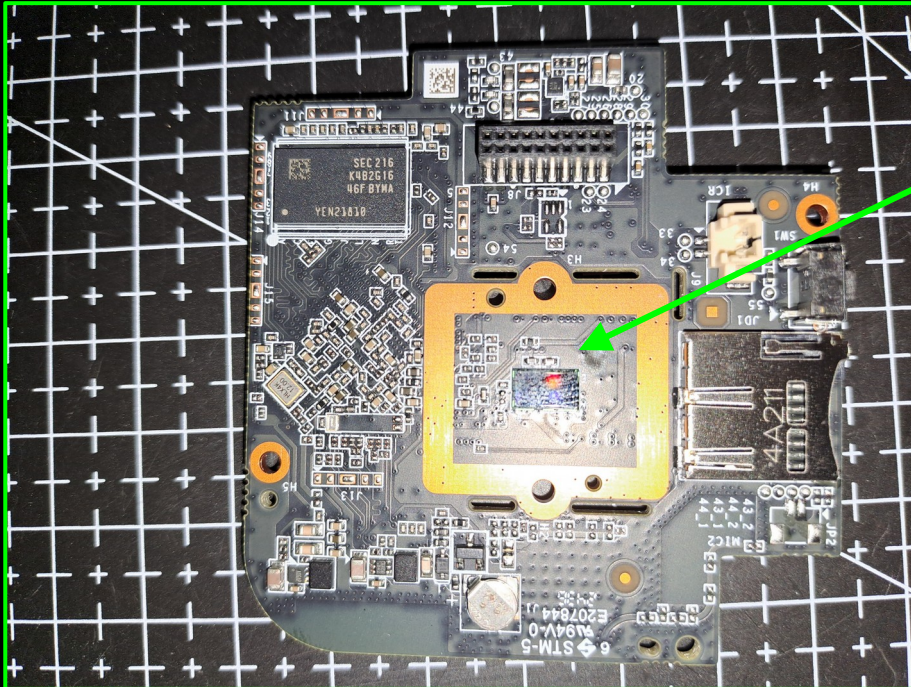
For the 1st board

Camera JST



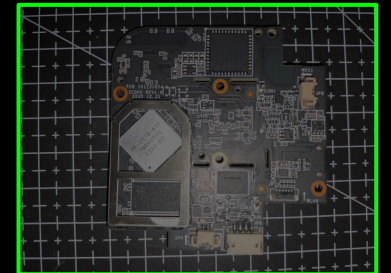
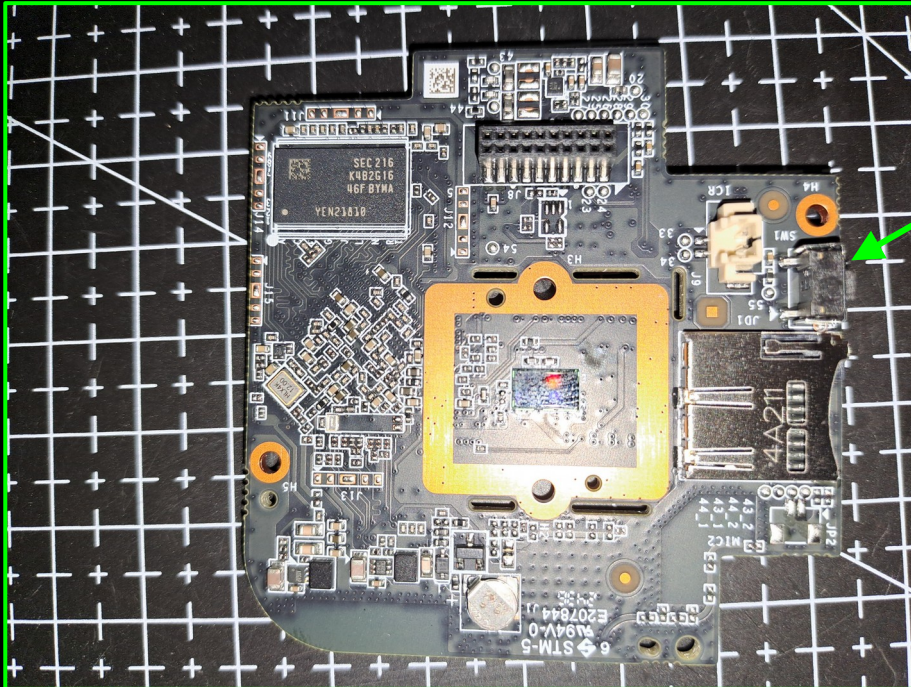
camera

Camera Footprint



buttons

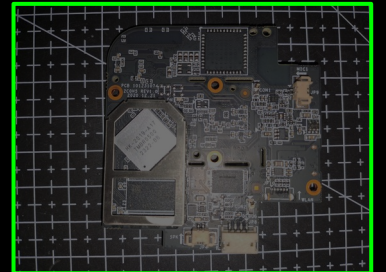
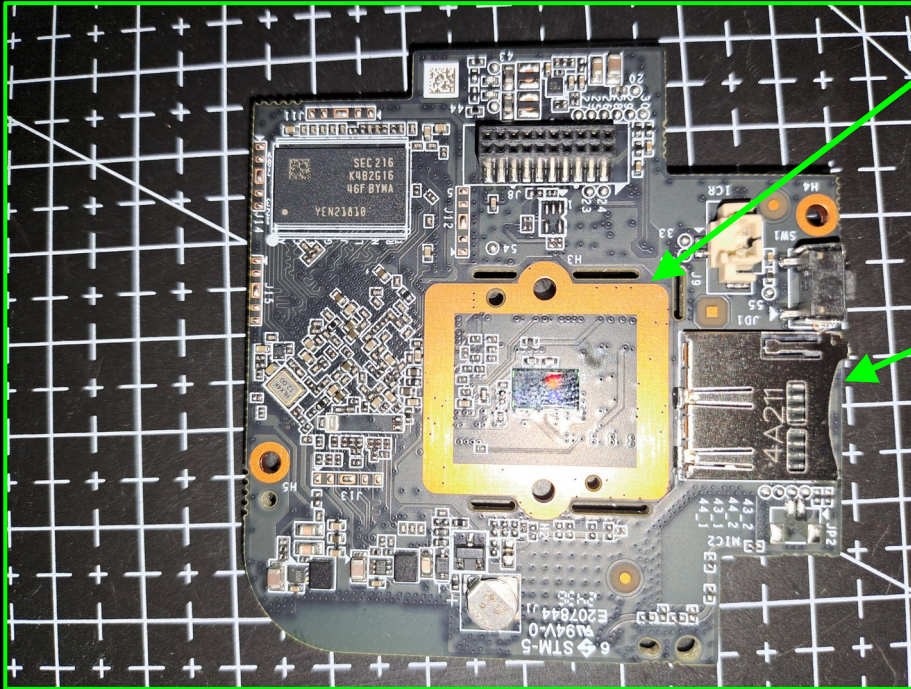
Reset



shielding & ground

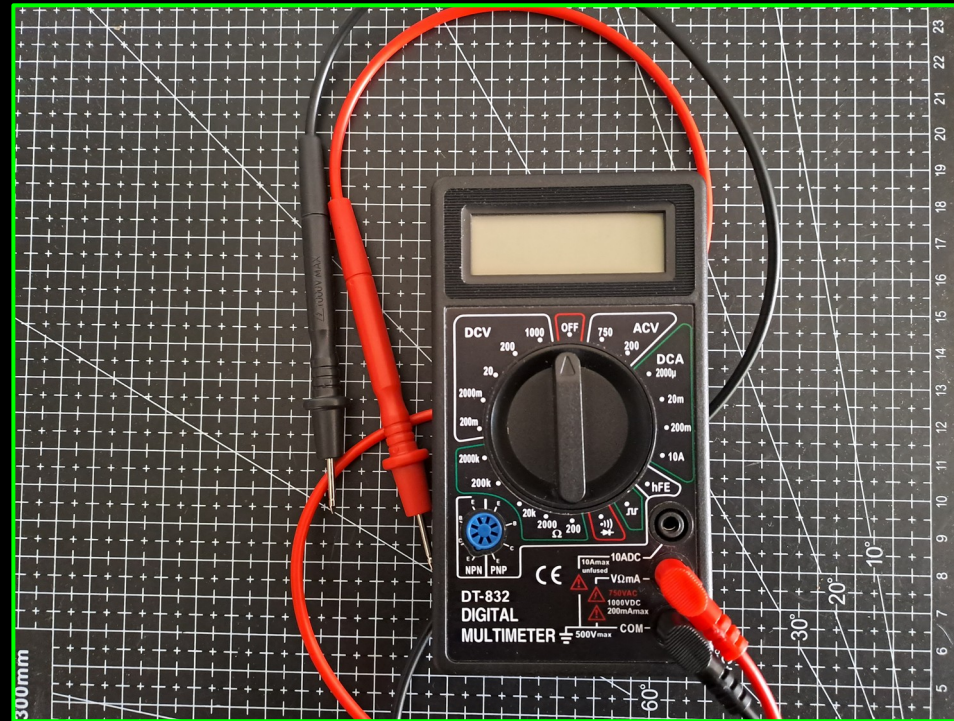
GND

SD Card Slot



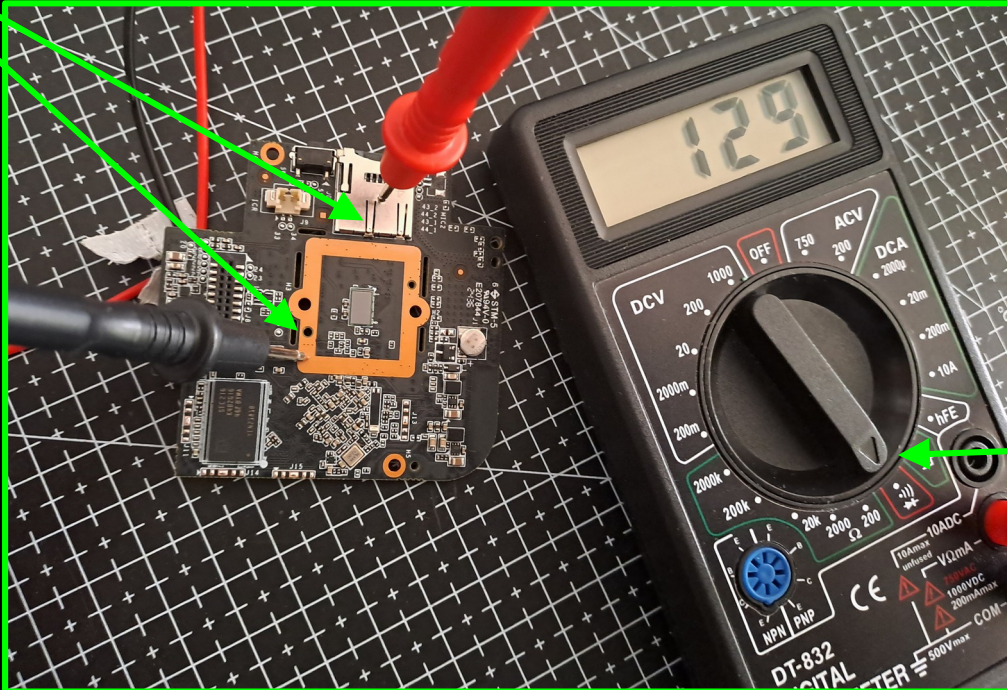
multimeter

- [*] Voltage
- [*] Current
- [*] Resistance
- [*] Continuity
- [*] more...



testing 2 points

probes

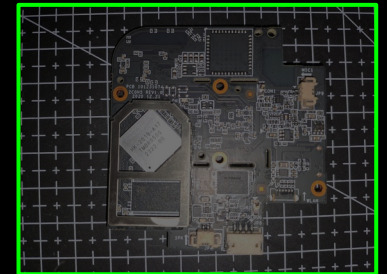
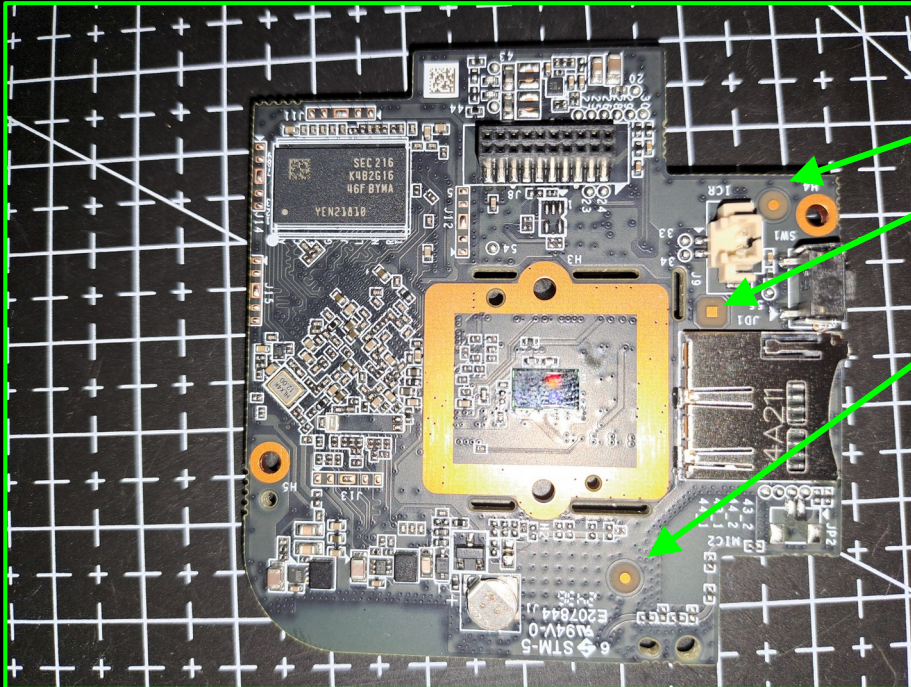


Continuity
mode

pins & pads

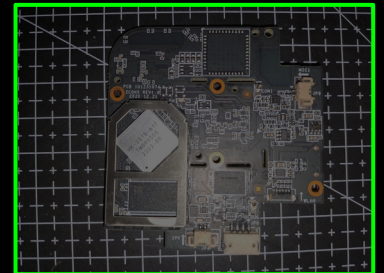
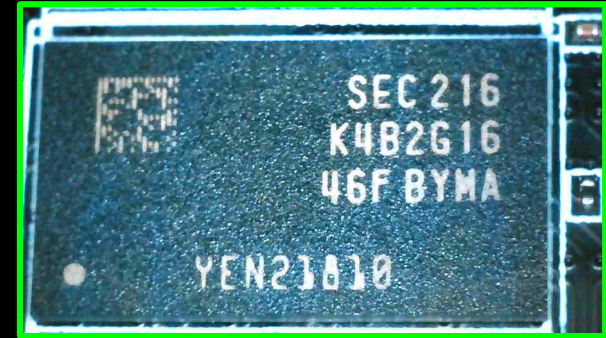
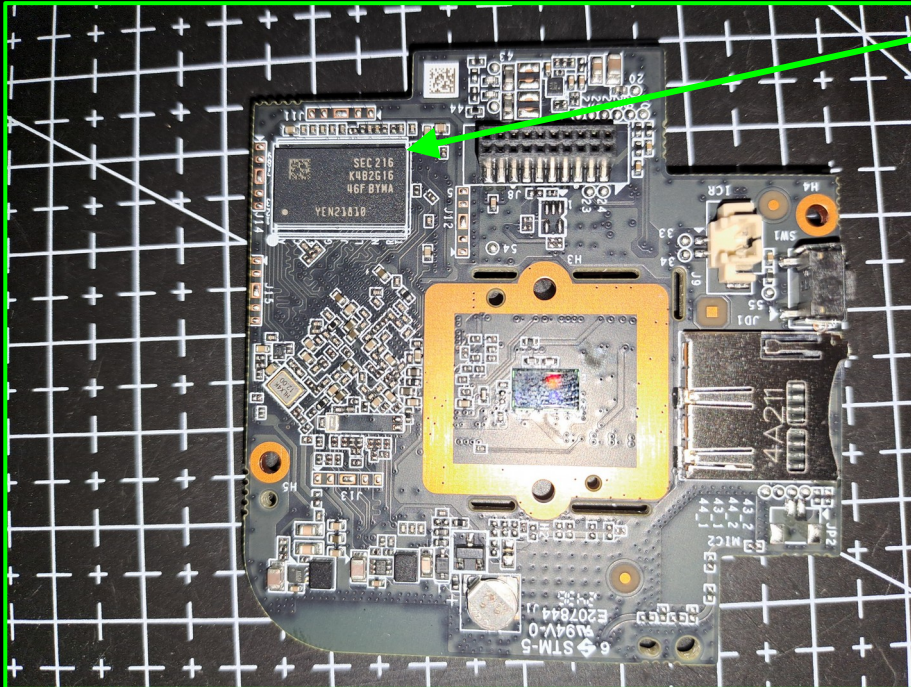
Pads

- [*] Oscilloscope
- [*] Logic Analyzer
- [*] Protocol Analyzer



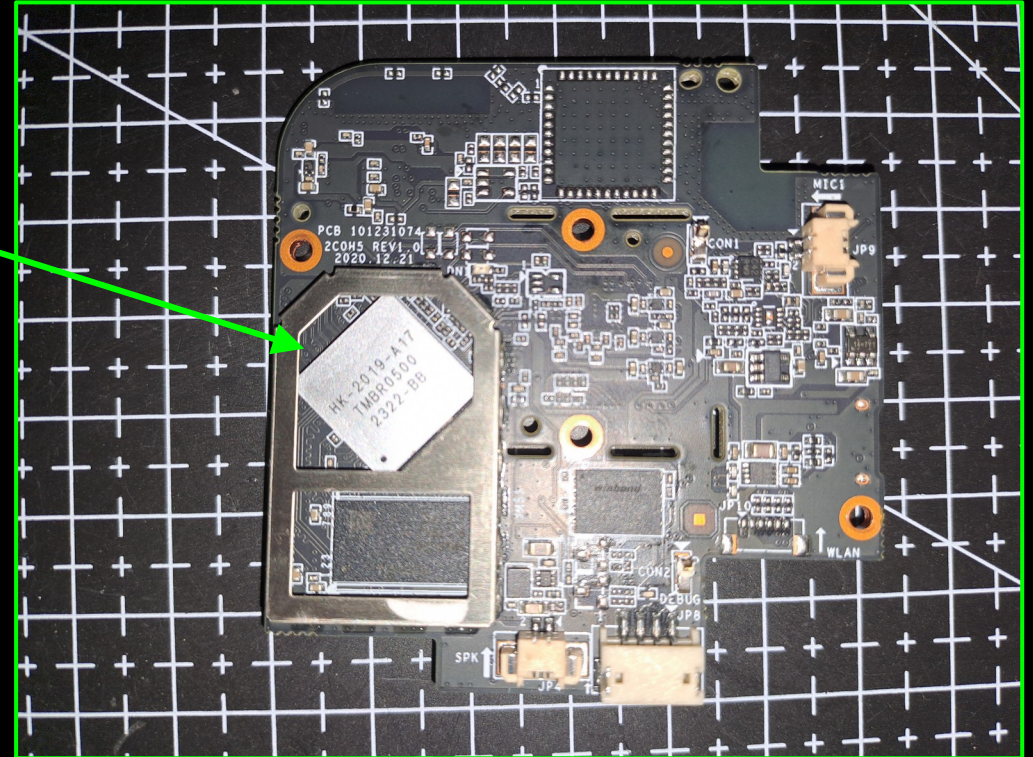
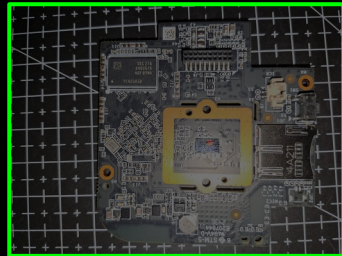
RAM

Samsung SDRAM



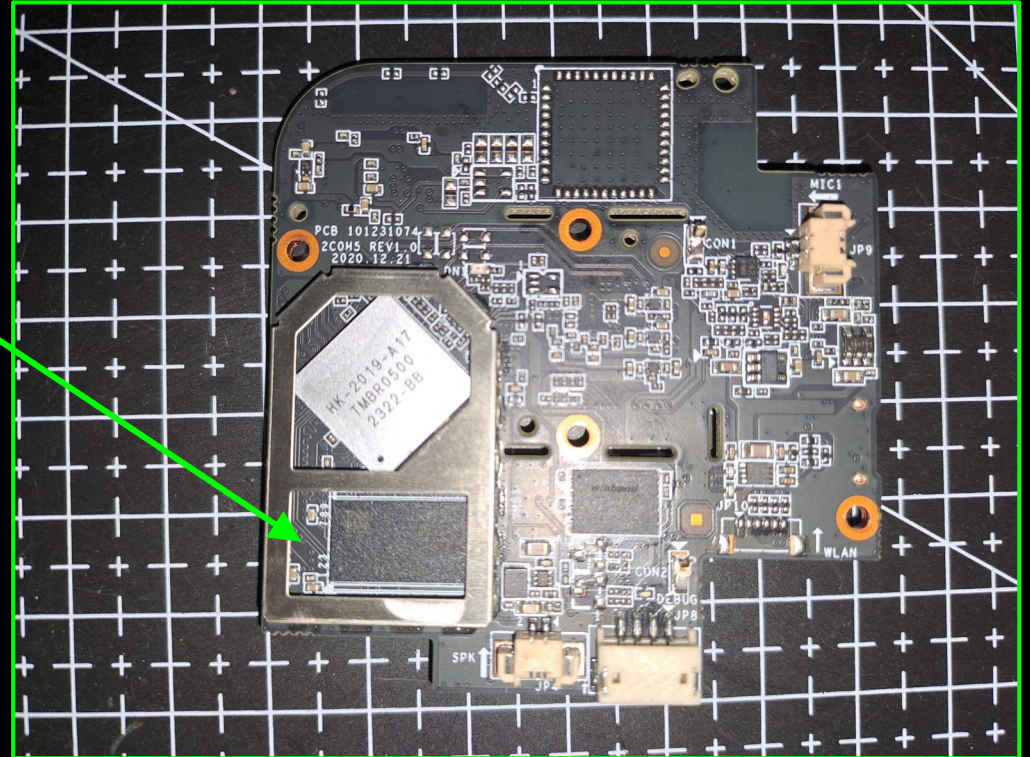
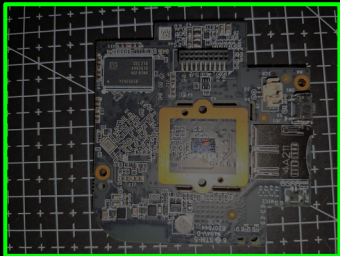
SoC

ARM Cortex



RAM

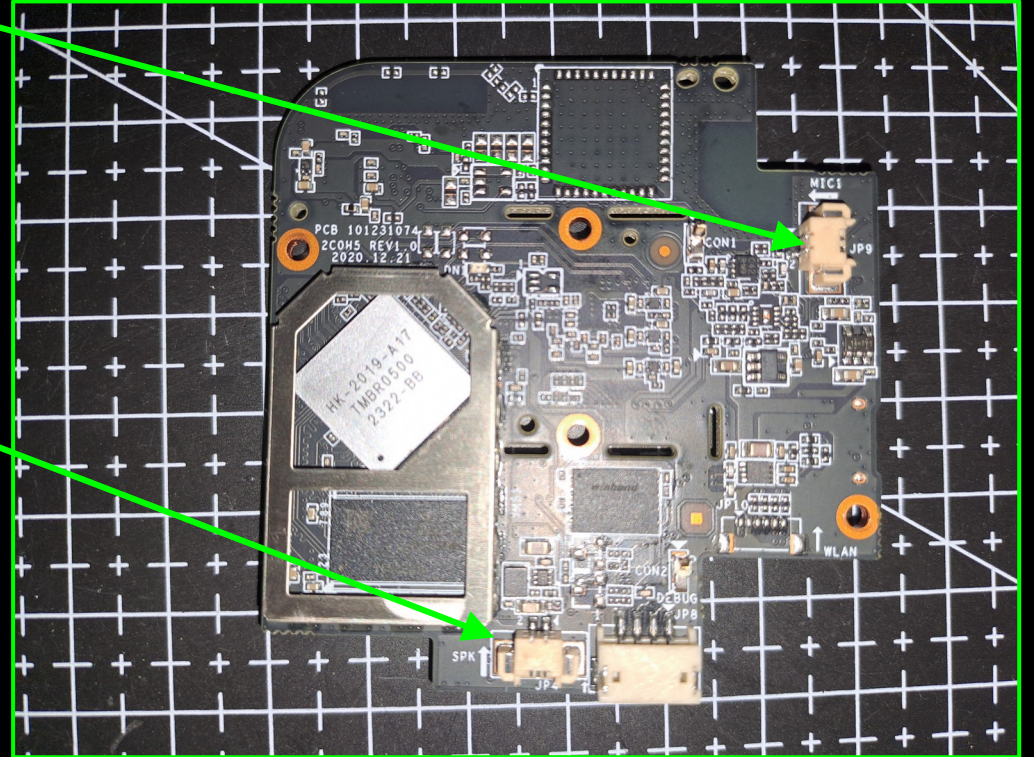
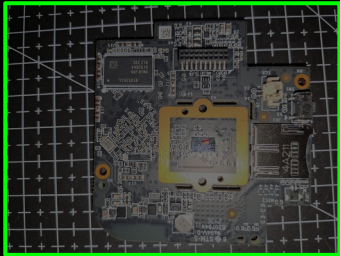
Samsung SDRAM



connectors

Microphone

Speaker

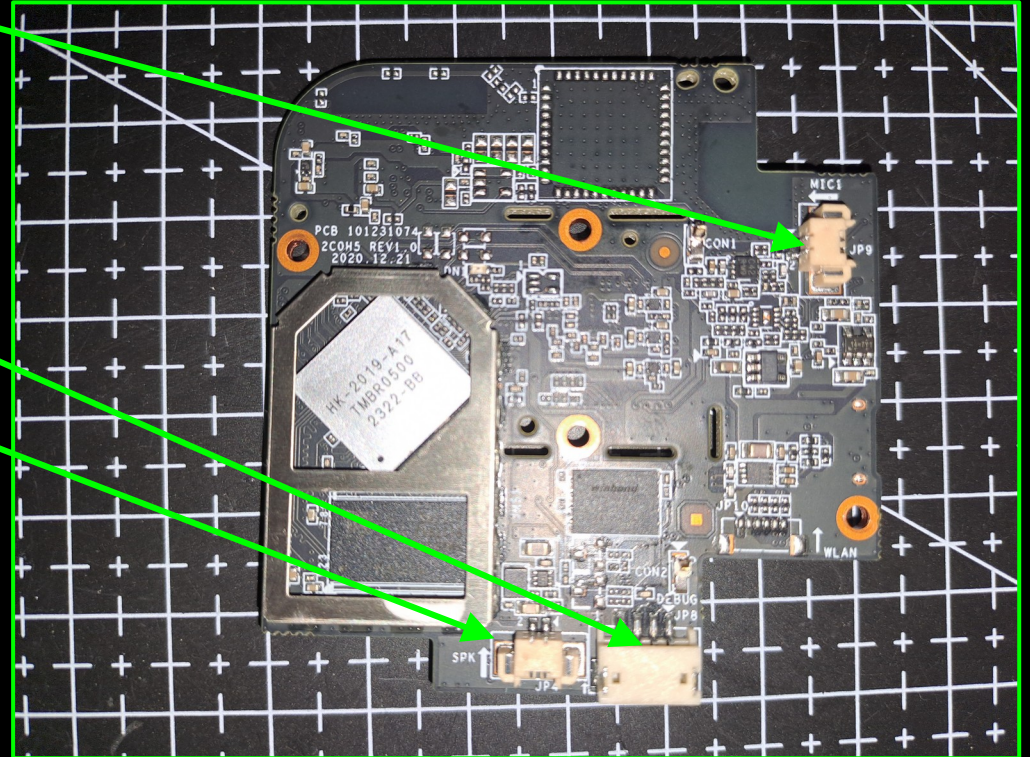
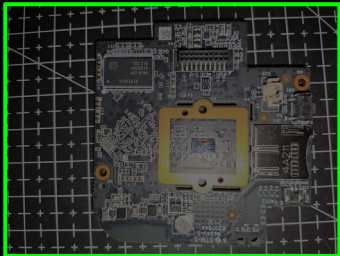


connectors

Microphone

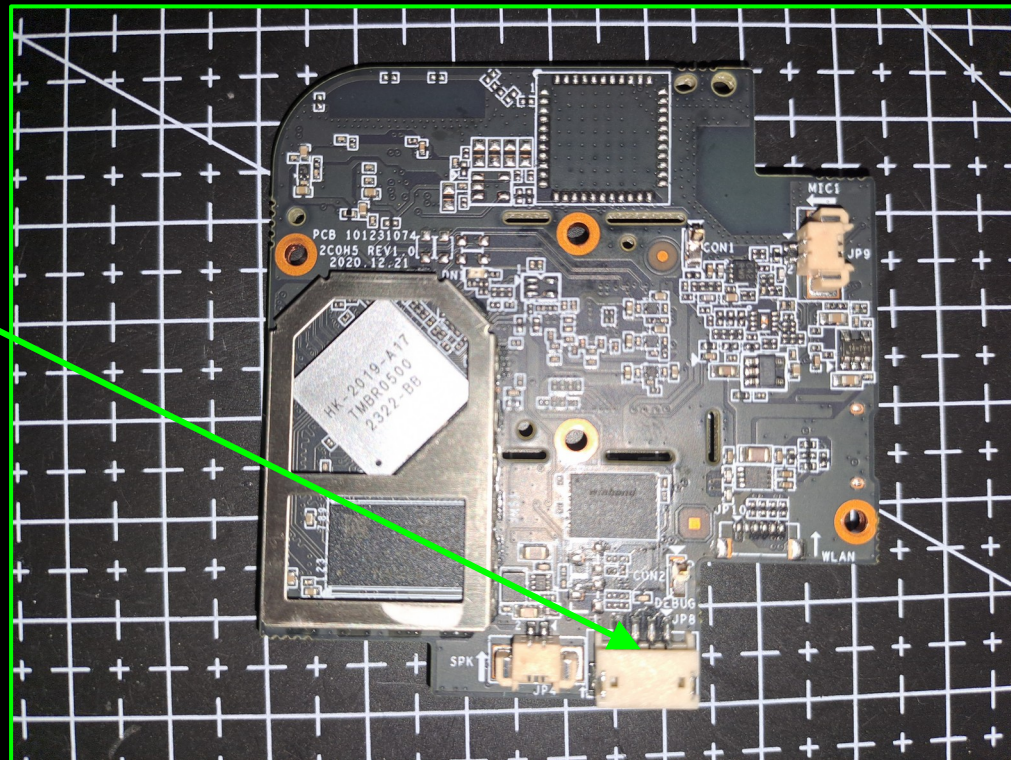
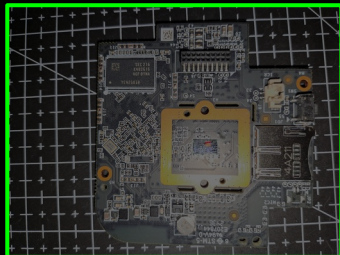
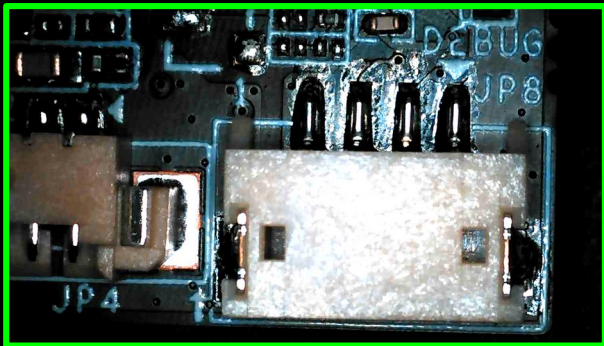
????????

Speaker



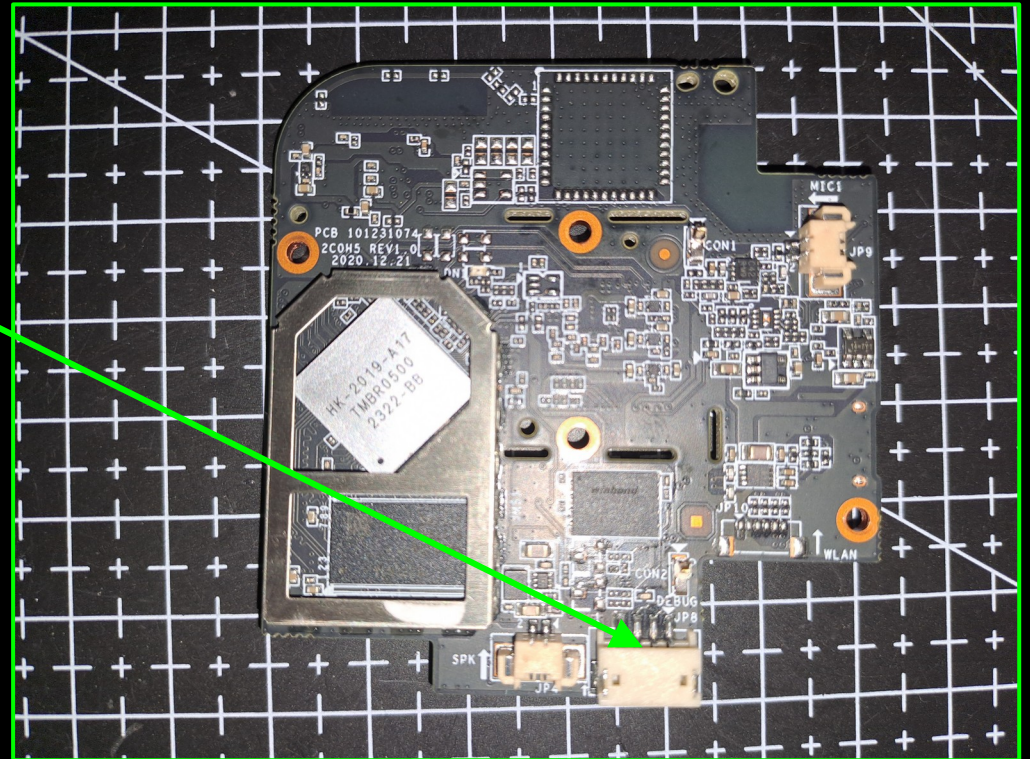
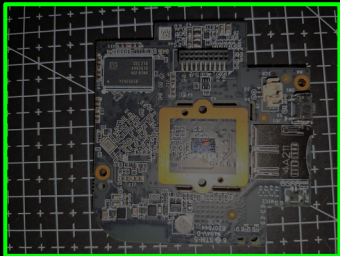
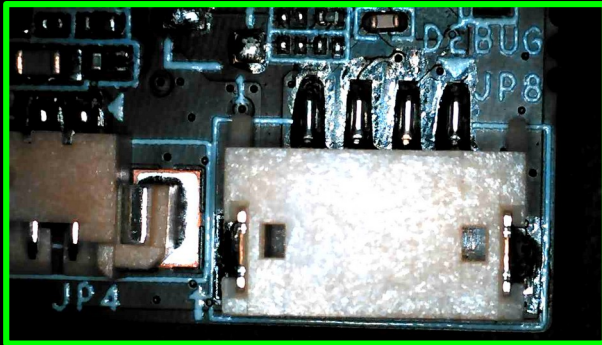
debug

DEBUG? JP8?



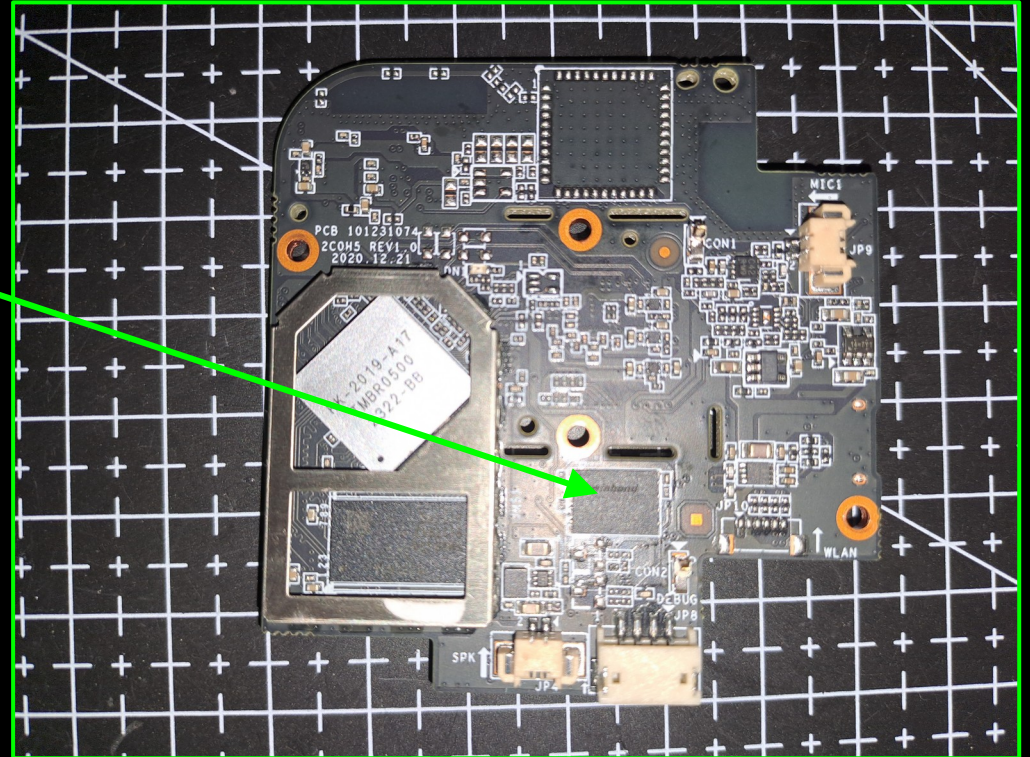
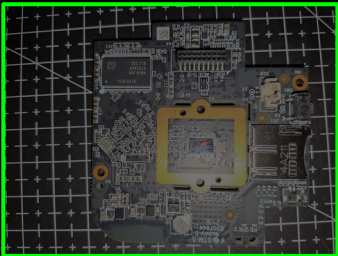
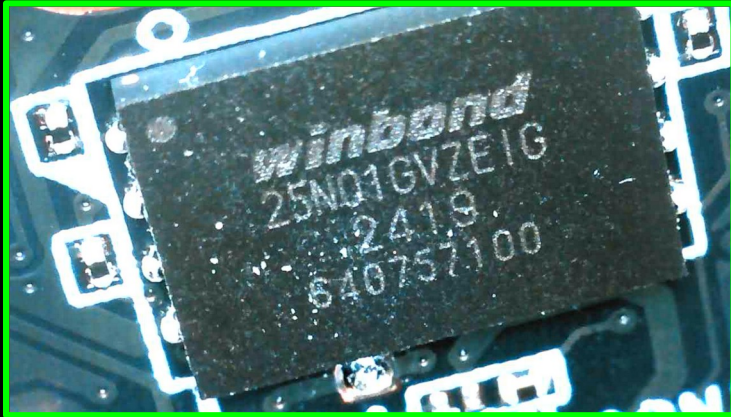
uart

[*] access to the OS



flash

Winbond ROM



summary

- [-] SoC (**S**ystem **o**n a **C**hip)
 - [*] ARMv7 architecture
 - [*] Most likely custom made
- [+] ROM (**R**ead **O**nly **M**emory)
- [+] UART (**U**niversal **A**synchronous **R**eceiver/**T**ransmitter)

summary

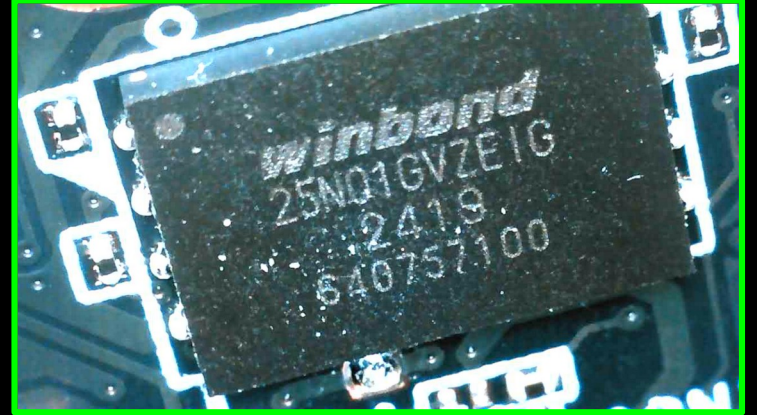
- [+] SoC (**S**ystem **o**n a **C**hip)
- [-] ROM (**R**ead **O**nly **M**emory)
 - [*] Winbond is the manufacturer
 - [*] It can store 128M data
 - [*] Has our 'entire' firmware
 - [*] No security features can be found
- [+] UART (**U**niversal **A**synchronous **R**eceiver/**T**ransmitter)

summary

- [+] SoC (**S**ystem **o**n a **C**hip)
- [+] ROM (**R**ead **O**nly **M**emory)
- [-] UART (**U**niversal **A**synchronous **R**eceiver/**T**ransmitter)
 - [*] Can be used to access the OS shell
 - [*] Usually locked
 - [*] 115200 baudrate (by default)

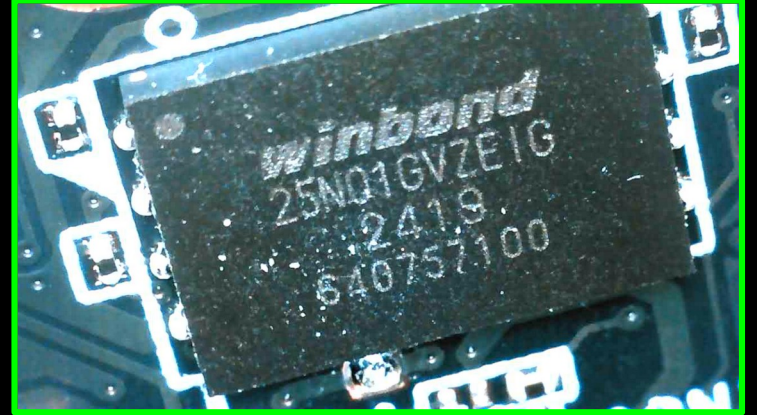
spi flash

[*] 128M storage size
[*] SPI protocol



spi flash

- [*] 128M storage size
- [*] SPI protocol
- [*] WSON8 package



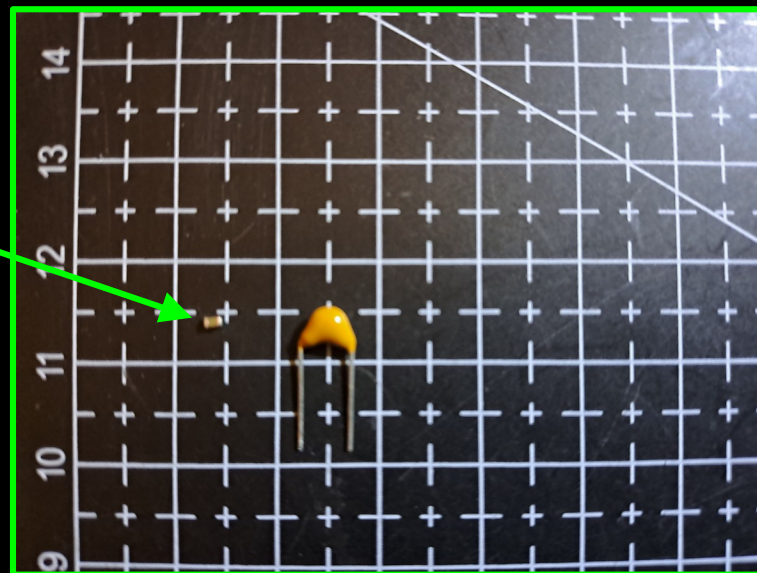
package types

[+] THT (Through-Hole Technology)



package types

- [+] THT (Through-Hole Technology)
- [+] SMT (Surface Mount Technology)

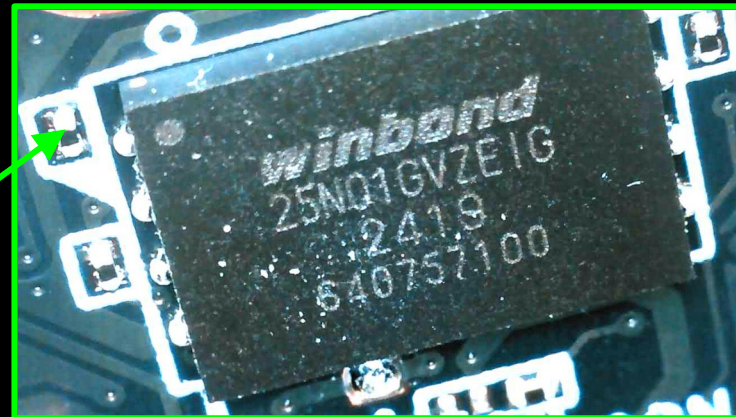


package types

- [+] THT (Through-Hole Technology)
- [-] SMT (Surface Mount Technology)
 - [*] SOP (Small Outline Package)
 - [*] QFN (Quad Flat No-lead)
 - [*] BGA (Ball Grid Array)
 - [*] WSON (Wettable-flank Small Outline No-lead)
 - [*] ...

spi flash

- [*] 128M storage size
- [*] SPI protocol
- [*] WSON8 package
- [*] Small components can be found around it



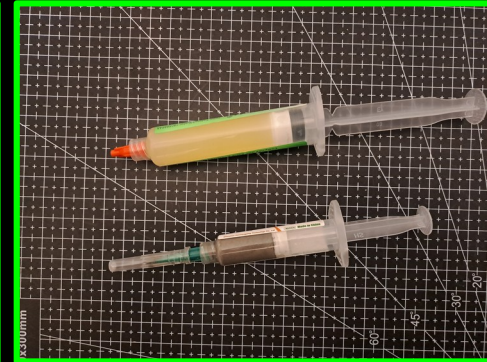
desoldering

- [*] provides true isolation
- [*] its very tedious
- [*] special packages (e.g.: BGA) are not easy to solder



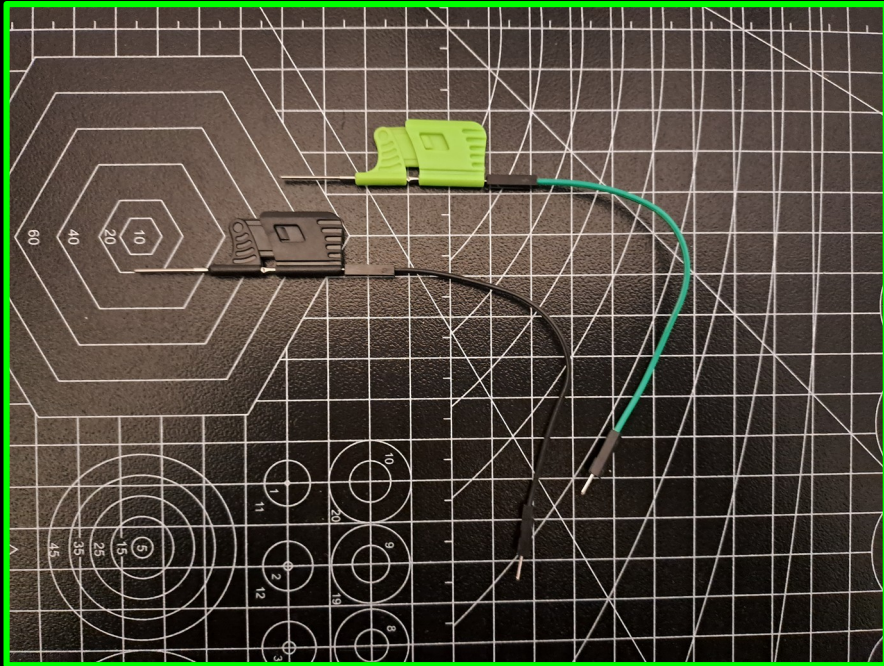
desoldering

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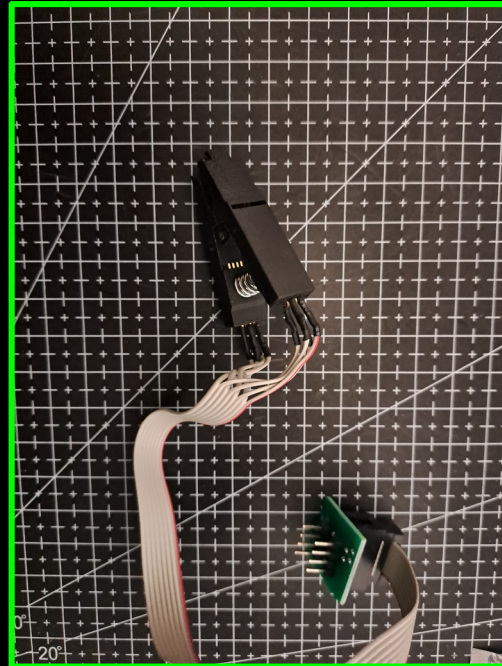
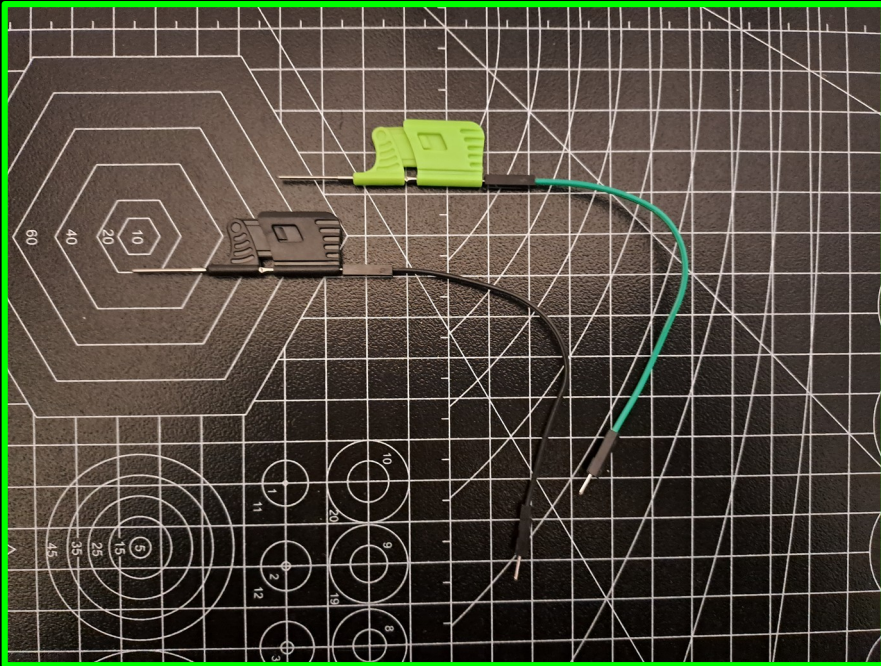
probing

- [*] quick and easy
- [*] does not work with BGA packages
- [*] no protection for reverse current



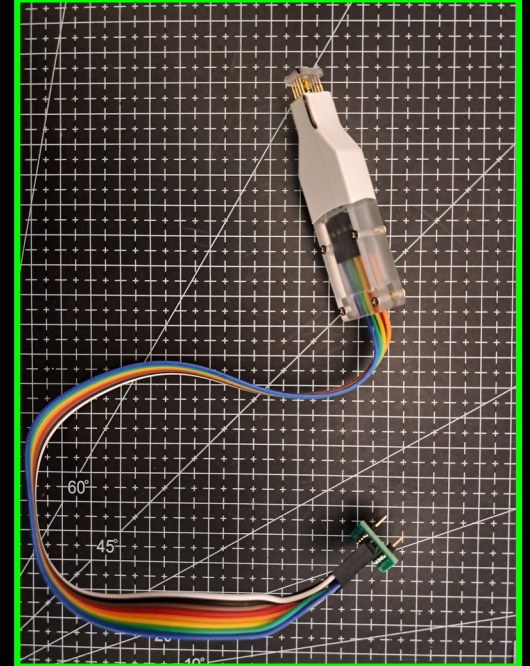
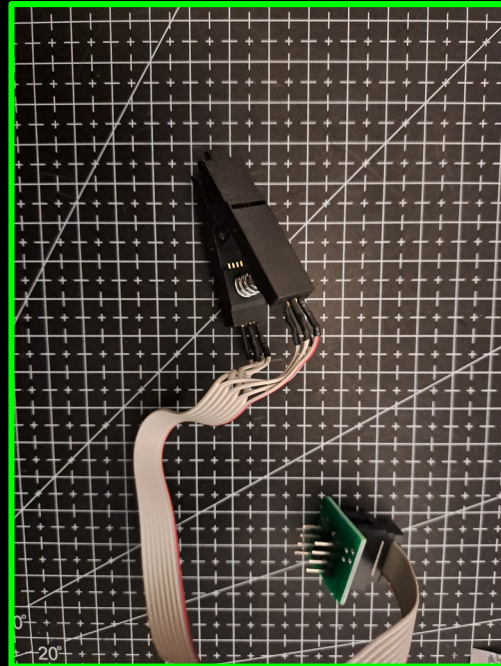
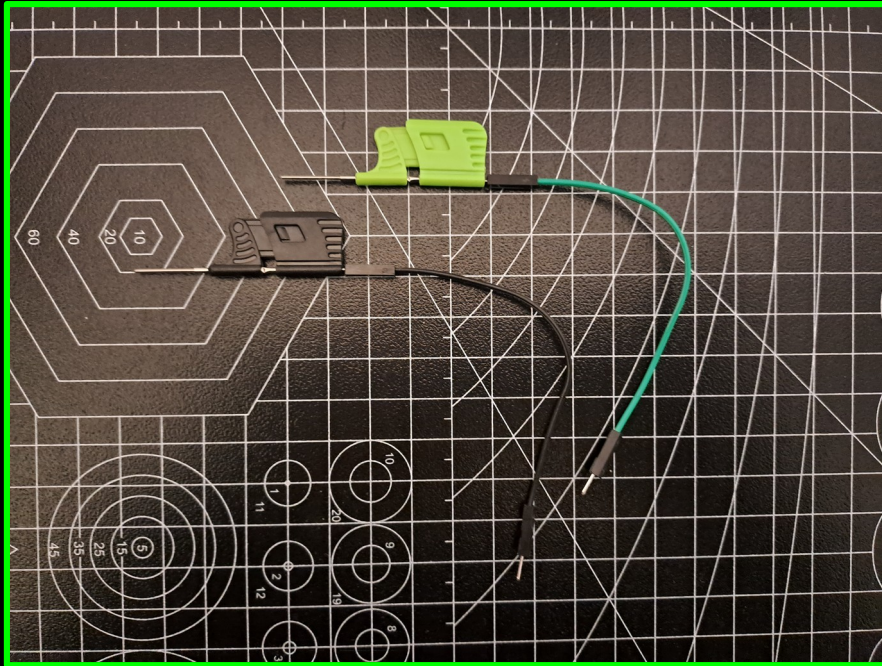
probing

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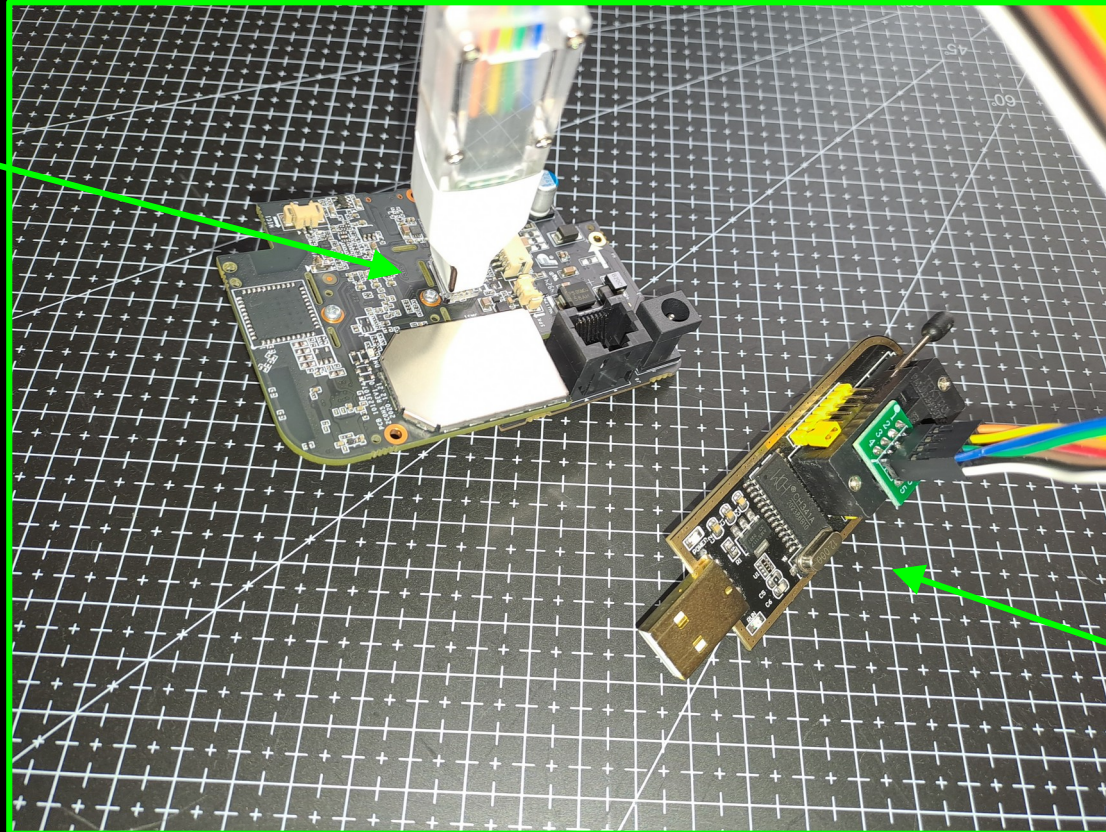
probing

- [*] quick and easy
- [*] does not work with BGA packages
- [*] no protection for reverse current



sbi setup

SPI Flash



CH341A

spi reader software

[flashrom]	will address 99% of cases
[snander]	will cover 0.99% of cases
[custom]	for 0.01%, use your own MCU

snander

```
$ snander -r firmware
```


snander

```
$ snander -r firmware
$ snander -r firmware.2
```

snander

```
$ snander -r firmware
```

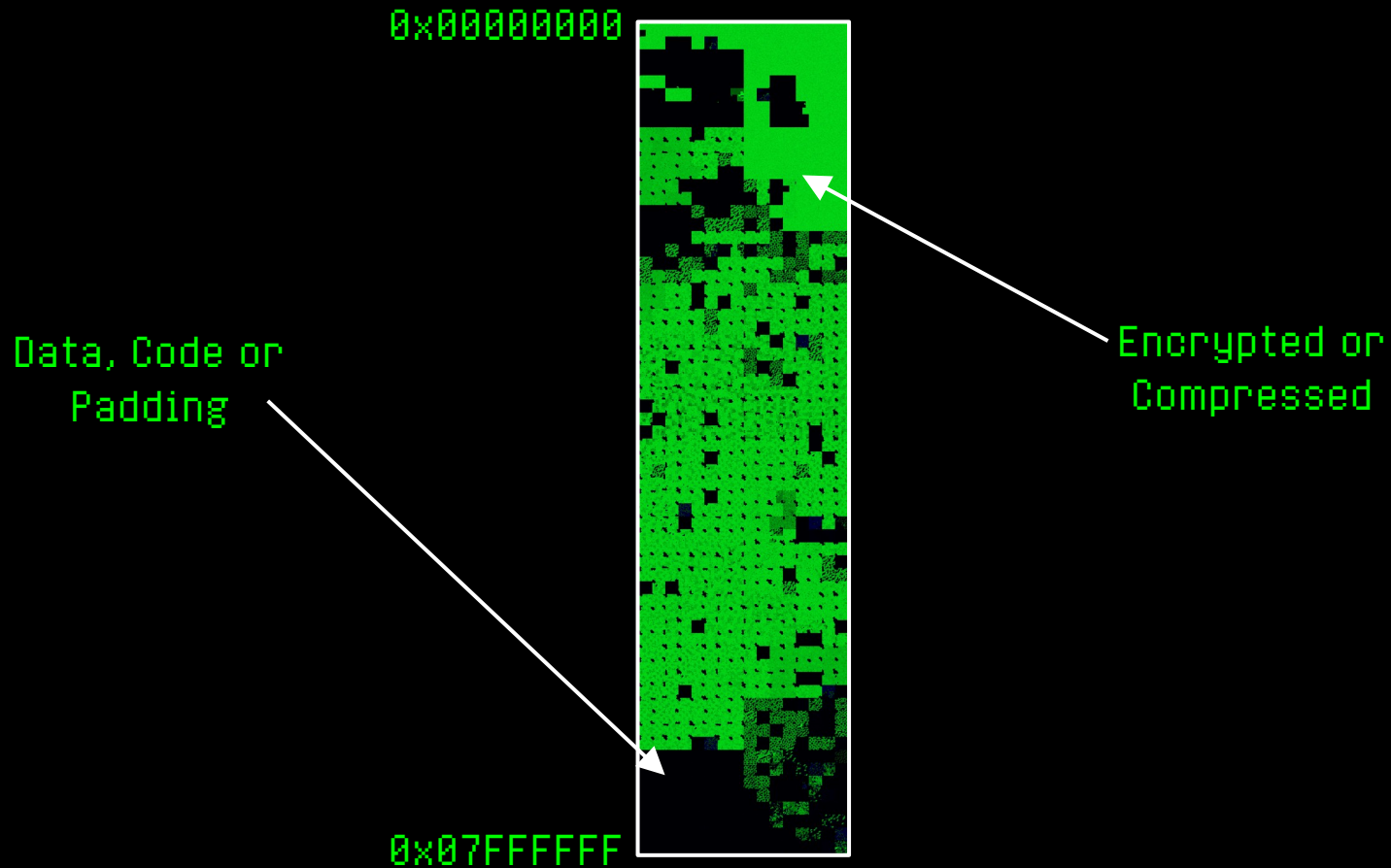
```
$ snander -r firmware.2
```

```
$ md5sum firmware*
```

```
b2048ed3f684ae33458e440a86252774  firmware
```

```
b2048ed3f684ae33458e440a86252774  firmware.2
```

entropy



binwalk

```
$ binwalk -Me firmware
```

```
-----  
HEXADECIMAL      DESCRIPTION  
-----  
0x07000000      Device tree blob (DTB)  
0x0F000000      Device tree blob (DTB)  
0x17000000      UBI image, version: 1  
0x6B000000      UBI image, version: 1  
0x73400000      UBI image, version: 1  
0x76000000      UBI image, version: 1  
...  
0x78C00000      UBI image, version: 1  
0x79000000      UBI image, version: 1  
-----
```


device tree

```
$ grep "nand" system.dts
```

```
nand@f0400000 {  
    ...  
    partition_bld {  
        label = "bld";  
        reg = <0x00 0x00 0x00 0x200000>;  
    };  
    partition_teeos0 {  
        label = "teeos0";  
        reg = <0x00 0x200000 0x00 0x180000>;  
    };  
};
```

firmware structure



enviornment partition

```
$ dd if=firmware skip=$((0x500000)) bs=1 | hexdump -C | head
00000000  52 e2 b8 71 61 72 63 68 3d 61 72 6d 00 62 61 75 |R..qarch=arm.baul
00000010  64 72 61 74 65 3d 31 31 35 32 30 30 00 62 6f 61 |ldrate=115200.boal
00000020  72 64 3d 6e 76 74 2d 6e 61 35 31 30 35 35 00 62 |lrd=nvt-na51055.bl
00000030  6f 61 72 64 5f 6e 61 6d 65 3d 6e 76 74 2d 6e 61 |load_name=nvt-na1
00000040  35 31 30 35 35 00 62 6f 6f 74 61 72 67 73 3d 65 |51055.bootargs=el
00000050  61 72 6c 79 70 72 69 6e 74 6b 20 63 6f 6e 73 6f |arlyprintk consol
00000060  6c 65 3d 74 74 79 53 30 2c 31 31 35 32 30 30 20 |le=ttyS0,115200 l
00000070  72 6f 6f 74 77 61 69 74 20 6e 70 72 6f 66 69 6c |rootwait nprofill
00000080  65 5f 69 72 71 5f 64 75 72 61 74 69 6f 6e 3d 6f |le_irq_duration=ol
00000090  6e 20 72 6f 6f 74 3d 75 62 69 30 3a 72 6f 6f 74 |n root=ubi0:rootl
```

enviornment partition

```
qarch=arm
board=nvt-na51055
board_name=nvt-na51055
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
          root=ubi0:rootfs rootfstype=ubifs ubi
          fm_autoconvert=1 init=/linuxrc boot_mode=0
bootcmd=loadk;bootm
bootdelay=3
cpu=armv7
dbg=0
ethact=eth_hikethaddr=08:cc:81:61:fd:15
fdtcontroladdr=6f9c600
gatewayip=192.168.1.254
hostname=soclnx
ipaddr=192.168.1.71
netmask=255.255.255.0
serverip=192.168.1.128
verify=0
```


arm trusted firmware

Reset

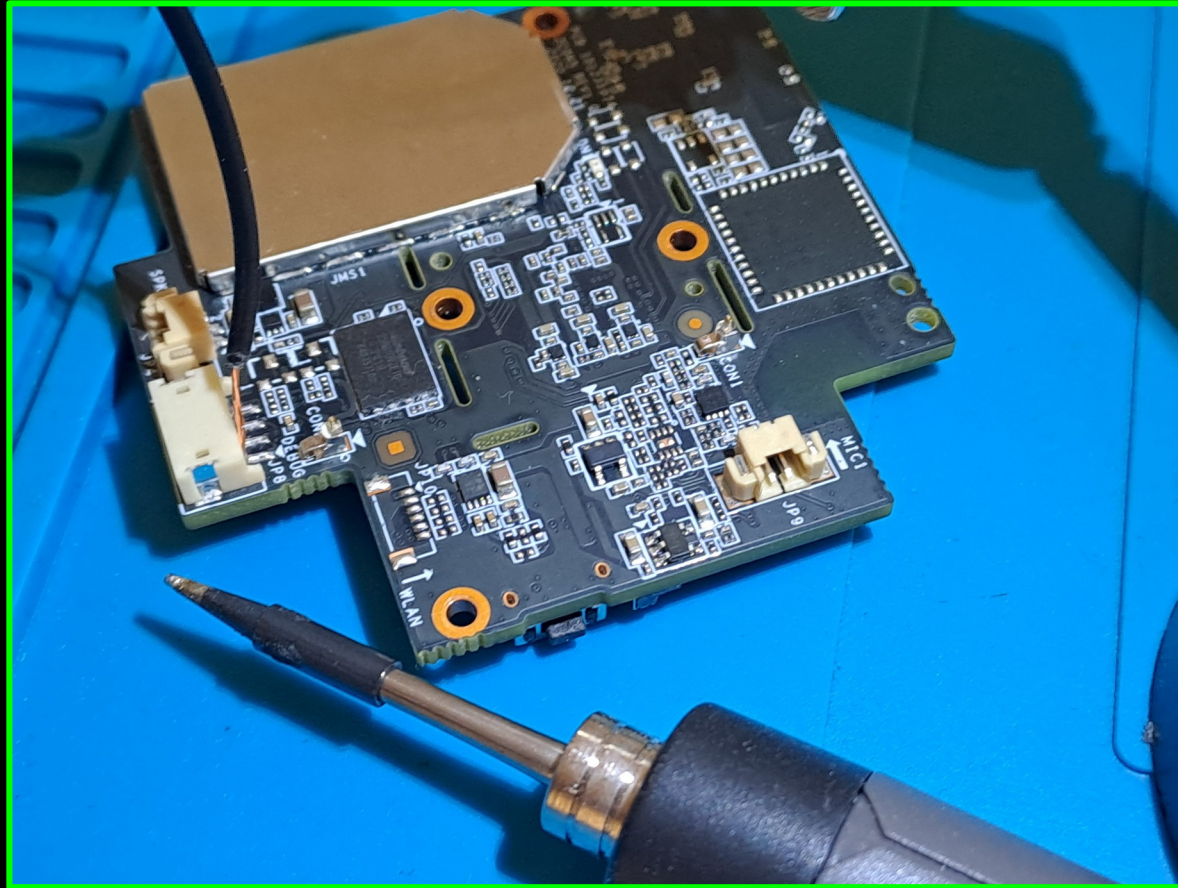
Boot Code [FUSE]

U-Boot/TEEOS [ROM]

TEEOS/U-Boot [ROM]

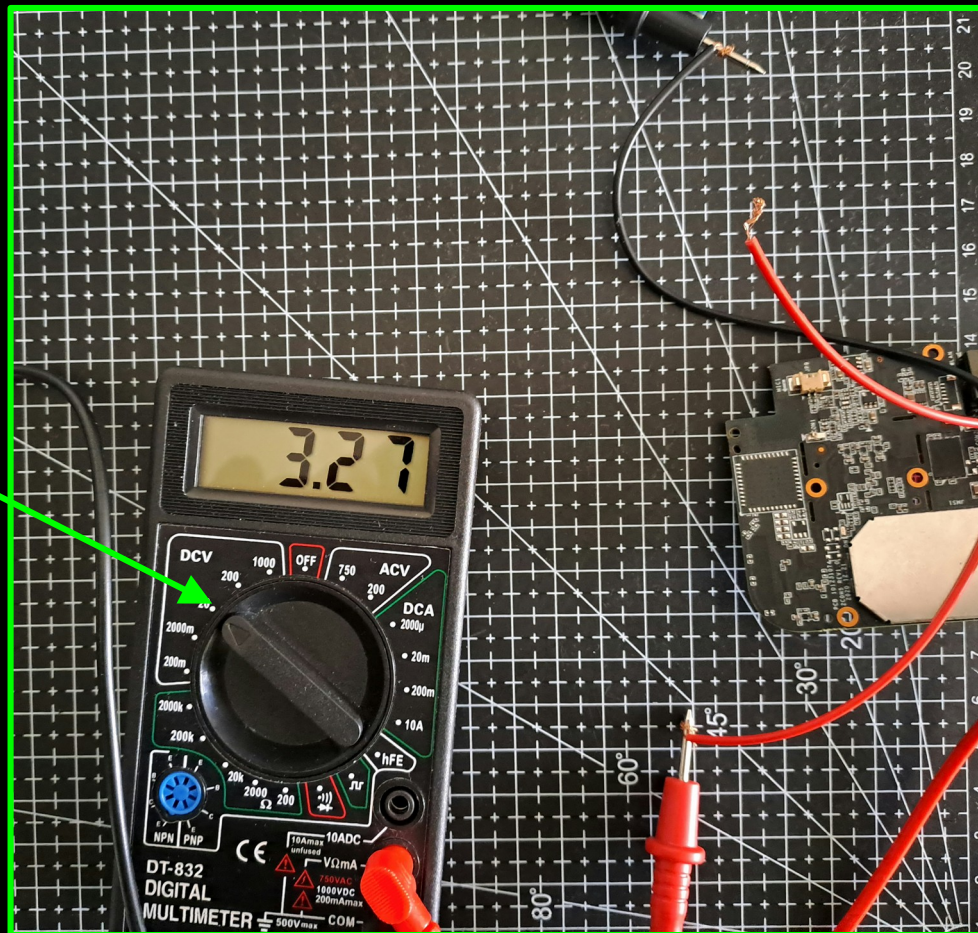
Linux [ROM]

soldering wires

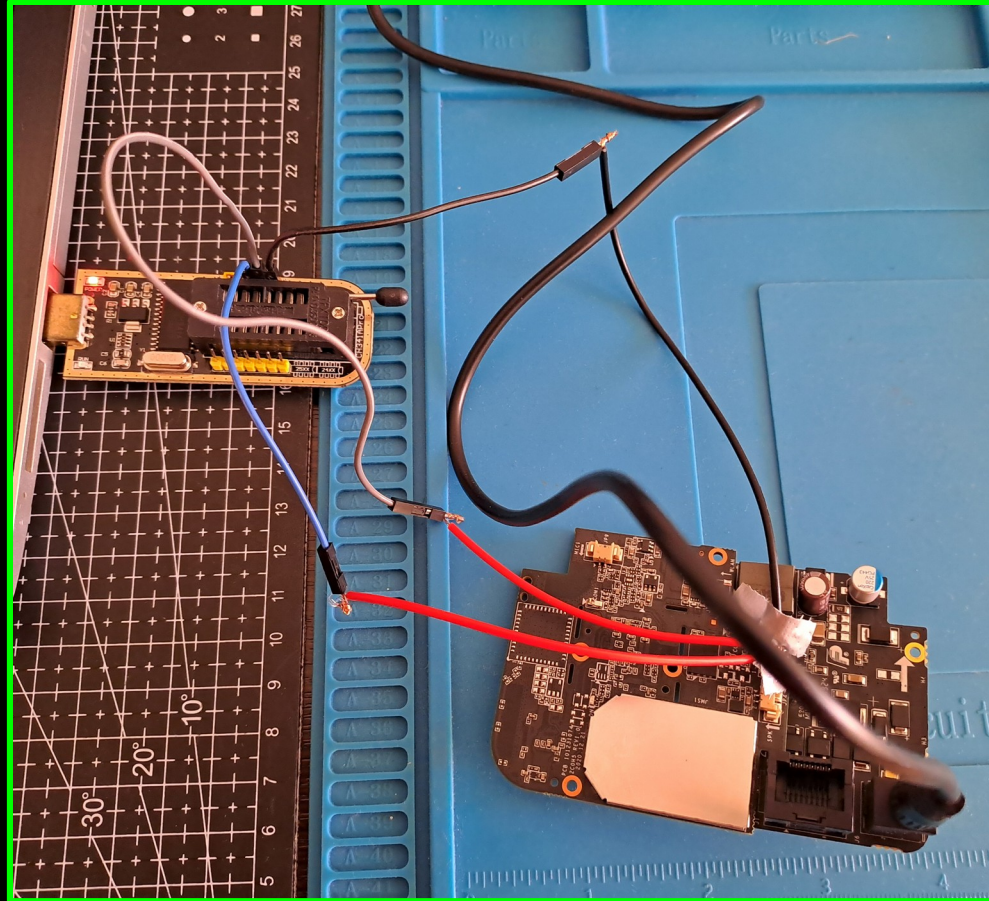


measuring voltage

Voltage Mode



uart setup



serial console

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
[Uboot] In release mode!
```

```
Hit Ctrl+u to stop autoboot:  0
```

```
eth_hik Waiting for PHY auto negotiation to complete..... TIMEOUT !
```

```
...
```

```
TFTP server auto connect disabled
```

```
booting from sys part...
```

```
#
```

u-boot shell

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
HKUS# printenv
```

```
arch=arm
```

```
baudrate=115200
```

```
board=nvt-na51055
```

```
board_name=nvt-na51055
```

```
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
```

```
root=ubi0:rootfs rootfstype=ubifs ubi.fm_autoconvert=1 init=/linuxrc
```

```
boot_mode=0
```

```
bootcmd=loadk;bootm
```

```
bootdelay=3
```

```
cpu=armv7
```

```
dbg=0
```

```
ethact=eth_hik
```

```
...
```

replacing boot variables

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
HKUS# env set bootdelay 10
```

```
HKUS# env set bootargs "... init=/bin/sh ..."
```

```
HKUS# printenv
```

```
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
```

```
root=ubi0:rootfs rootfstype=ubifs ubi.fm_autoconvert=1 init=/linuxrc
```

```
boot_mode=0
```

```
bootdelay=10
```

reset the SoC

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
HKUS# env set bootdelay 10
```

```
HKUS# env set bootargs "... init=/bin/sh ..."
```

```
HKUS# printenv
```

```
bootargs=earlyprintk console=ttyS0,115200 rootwait nprofile_irq_duration=on
```

```
root=ubi0:rootfs rootfstype=ubifs ubi.fm_autoconvert=1 init=/linuxrc
```

```
boot_mode=0
```

```
bootdelay=10
```

```
HKUS# saveenv
```

```
HKUS# reset
```


patching the firmware

```
$ diff <(hexdump -C firmware.0) <(hexdump -C firmware)
< 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 6c lonvert=1 init=/1|
< 005000d0 69 6e 75 78 72 63 20 20 20 62 6f 6f 74 5f 6d 6f linuxrc boot_mol
---
> 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d ?? ?? lonvert=1 init=??|
> 005000d0 ?? ?? ?? ?? ?? ?? 20 20 20 62 6f 6f 74 5f 6d 6f |?????? boot_mol
```

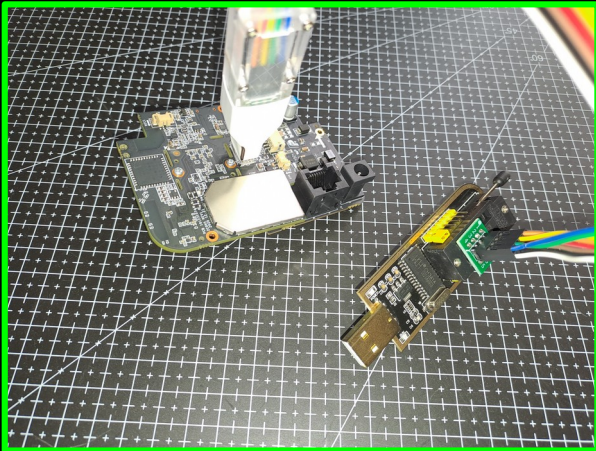
patching the firmware

```
$ diff <(hexdump -C firmware.0) <(hexdump -C firmware)
< 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 6c lonvert=1 init=/ll
< 005000d0 69 6e 75 78 72 63 20 20 20 62 6f 6f 74 5f 6d 6f linuxrc boot_mol
---
> 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 62 lonvert=1 init=/bl
> 005000d0 69 6e 2f 73 68 20 20 20 20 62 6f 6f 74 5f 6d 6f lin/sh boot_mol
```

patching the firmware

```
$ diff <(hexdump -C firmware.0) <(hexdump -C firmware)
< 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 6c lonvert=1 init=/l
< 005000d0 69 6e 75 78 72 63 20 20 20 62 6f 6f 74 5f 6d 6f linuxrc boot_mol
---
> 005000c0 6f 6e 76 65 72 74 3d 31 20 69 6e 69 74 3d 2f 62 lonvert=1 init=/bl
> 005000d0 69 6e 2f 73 68 20 20 20 20 62 6f 6f 74 5f 6d 6f lin/sh boot_mol

$ snander -w firmware.0
```



unlocked firmware

```
$ picocom -b 115200 /dev/ttyUSB0
```

```
...
```

```
#
```

questions?

[Repository] <https://github.com/0x0000z3r0/hikvision-re>
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?



Github



LinkedIn

